

ZHANGYE, China

WMO: 526520

Lat: 39.0795N

Lon: 100.2750E

Elev: 1462

StdP: 84.95

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.0	-17.7	-26.1	0.4	-13.1	-24.2	0.5	-11.2	7.7	-4.9	6.5	-4.1	1.5	200	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	33.4	17.0	31.7	16.4	30.2	15.8	19.5	27.9	18.5	26.9	17.7	25.9	2.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	14.7	22.0	16.1	13.7	21.0	15.1	12.8	20.7	62.7	27.8	58.9	26.8	56.0	26.0	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.6	6.3	5.3	DB	-22.8	36.6	2.7	1.4	-24.7	37.6	-26.3	38.4	-27.9	39.2	-29.9	40.3	
			WB	-23.1	21.0	2.7	1.4	-25.0	22.0	-26.6	22.8	-28.1	23.6	-30.0	24.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-8.4	-3.5	3.7	11.4	16.8	21.2	23.2	21.8	16.6	9.1	1.0	-6.4
	DBStd	11.53	4.16	4.40	4.51	4.39	3.59	2.91	2.90	2.86	3.14	3.66	3.94	4.07
	HDD10.0	2021	569	377	198	35	2	0	0	0	1	62	270	507
	HDD18.3	3828	827	611	453	209	71	9	1	5	69	287	520	766
	CDD10.0	1635	0	0	4	78	212	335	410	366	198	33	0	0
	CDD18.3	401	0	0	0	2	23	94	153	113	17	0	0	0
	CDH23.3	4340	0	0	1	57	352	1019	1586	1106	210	10	0	0
	CDH26.7	1590	0	0	0	11	84	352	693	407	42	1	0	0
Wind	WSAvg	2.4	2.0	2.4	2.7	2.9	2.7	2.5	2.4	2.3	2.1	2.1	2.1	2.1
Precipitation	PrecAvg	209	2	2	6	10	22	36	47	41	28	10	3	1
	PrecMax	257	7	4	17	17	35	64	75	76	51	18	7	3
	PrecMin	157	0	1	1	4	7	14	27	21	6	2	0	0
	PrecStd	25	1	1	4	4	7	12	13	12	11	4	2	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	13.5	21.2	28.2	31.2	34.0	36.3	34.7	30.3	24.7	16.9	9.7
		MCWB	0.6	2.9	6.8	10.5	13.3	16.0	18.5	18.1	14.9	10.4	5.6	1.4
	2%	DB	4.9	9.9	17.6	24.8	29.0	31.8	34.1	32.3	27.7	21.9	14.0	6.0
		MCWB	-1.7	1.2	5.5	9.4	12.4	15.1	17.7	17.0	13.6	9.6	4.4	-0.7
	5%	DB	1.8	7.2	15.1	22.3	27.0	30.2	32.3	30.7	25.7	19.5	11.6	3.4
		MCWB	-3.4	-0.3	4.4	8.5	11.6	14.7	17.2	16.4	13.2	8.6	3.2	-2.1
	10%	DB	-0.9	4.9	12.7	20.3	25.2	28.8	30.6	29.0	23.8	17.2	9.0	1.0
		MCWB	-5.1	-1.5	3.3	7.7	11.0	14.5	16.8	16.1	12.7	7.7	2.0	-3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	3.2	7.3	11.7	14.8	18.7	21.8	20.7	17.5	12.0	6.2	1.4
		MCDB	9.1	12.7	19.4	25.0	28.0	28.0	31.7	28.2	23.0	20.0	15.7	9.2
	2%	WB	-1.7	1.3	5.9	10.2	13.4	17.5	19.8	19.4	15.8	10.5	4.7	-0.5
		MCDB	4.5	9.4	16.2	22.6	25.6	26.2	28.6	27.2	22.4	19.3	13.2	5.7
	5%	WB	-3.4	-0.1	4.8	9.1	12.5	16.6	18.8	18.3	14.9	9.4	3.5	-2.0
		MCDB	1.6	6.6	14.3	20.6	24.1	25.4	27.5	26.4	21.7	17.9	11.1	3.1
	10%	WB	-5.0	-1.3	3.6	8.1	11.7	15.8	18.0	17.4	14.0	8.3	2.2	-3.4
		MCDB	-1.1	4.5	12.1	18.9	22.6	24.8	26.6	25.3	20.5	16.1	8.8	0.8
Mean Daily Temperature Range	5% DB	MDBR	14.2	14.4	13.9	13.3	13.1	12.6	12.1	12.3	12.9	14.6	13.8	13.5
		MCDBR	18.0	18.0	17.9	16.7	16.7	15.8	15.6	15.4	16.2	17.6	17.7	16.8
		MCWBR	12.4	11.0	8.9	6.8	6.0	5.3	5.1	4.9	5.2	7.3	9.5	11.2
	5% WB	MCDBR	17.6	17.0	16.9	14.9	14.5	12.8	12.7	12.1	11.8	15.8	16.9	16.2
		MCWBR	12.2	10.4	8.5	6.1	5.6	5.0	4.8	4.7	4.6	6.6	9.1	10.8
Clear-Sky Solar Irradiance	taub		0.310	0.363	0.435	0.486	0.492	0.490	0.476	0.470	0.440	0.395	0.342	0.307
	taud		2.201	2.059	1.887	1.796	1.847	1.924	2.034	2.037	2.050	2.071	2.142	2.215
	Ebn at Noon		844	837	809	794	798	800	807	801	797	790	791	817
	Edh at Noon		102	136	180	211	204	189	168	164	152	133	108	94
All-Sky Solar Radiation	RadAvg		2.78	3.74	4.95	5.97	6.59	6.55	6.29	5.71	4.90	4.17	3.15	2.54
	RadStd		0.17	0.15	0.30	0.32	0.27	0.40	0.40	0.40	0.32	0.22	0.15	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+145	+63
Regional (3 neighbors)	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	-99	+66	N/A

Nomenclature: See separate page